

Wenrui Fan

✉ wenrui.fan@sheffield.ac.uk ☎ +44 (0) 7572 857388 🌐 wenruifan.github.io 🗣 wenruifan
🎓 Google Scholar 🔗 LinkedIn 🐙 Hugging Face ORCID: 0009-0007-1394-0092

Education

- **PhD Candidate in Computer Science**
– University of Sheffield, UK 01.2024 – Present
- **MSc in Robotics**
– University of Sheffield, UK; Distinction 09.2021 – 09.2022
- **BSc in Flight Vehicle Design and Engineering**
– Beijing Institute of Technology, China; Distinction BSc thesis 09.2017 – 07.2021

Appointments

- **AI Research Engineer – Centre for Machine Intelligence, University of Sheffield** 09.2023 – Present
– Lead and contribute to multimodal AI research in neuroimaging, precision medicine, cancer, protein science, and molecular modelling.
– Develop models, datasets, evaluation protocols, interpretation methods, and reproducible research software.
– Coordinate technical work with clinical, biomedical, and interdisciplinary collaborators.
– Contribute to collaborative funding proposals and open research infrastructure.
- **Research Engineer, AI for Manufacturing – Chongqing Research Institute of HIT** 11.2022 – 07.2023
– Led machine-learning development for semiconductor wafer anomaly detection.
– Improved detection performance and computational efficiency.
– Contributed to foundation-model architectures for intelligent manufacturing.

Technical Skills

AI and machine learning: Python, PyTorch ecosystem, scikit-learn, Hugging Face; multimodal learning, foundation-model adaptation, vision–language pre-training, graph learning, and generative modelling.

Evaluation and analysis: Baseline design, independent-cohort validation, subgroup analysis, model interpretation, causal analysis, and benchmarking.

Research software engineering: Git and GitHub, Linux, Docker, automated testing, continuous integration, Python packaging, and FAIR workflows.

Selected Research Projects

- **Neuropathic Pain Drug-response Prediction – Code**
– Led the design and PyTorch implementation of a multimodal framework using resting-state fMRI time-series and functional-connectivity modalities.
– Adapted a foundation model pre-trained on UK Biobank data to small clinical cohorts.
– Designed baselines and evaluation across two clinical datasets.
– Conducted subgroup-specific interpretation and compared learned patterns with medical and biological findings and theories.
- **MeDSLIP: Medical Vision–Language Pre-training – Code | Model**
– Led model design, implementation, experimentation, and release for fine-grained medical image–text alignment.
– Evaluated zero-shot learning, transfer learning, and fine-tuning.
– Released the code and pre-trained model for external reproduction and reuse.
- **Parkinson’s Disease Risk-factor Analysis**
– Led the machine-learning work for a UK Biobank study of potential Parkinson’s disease risk factors.
– Combined predictive modelling, age- and sex-stratified analysis, model interpretation, and causal inference.
– Distinguished predictive contribution from causal evidence to support appropriately cautious conclusions.
- **AI-driven Multistate Protein Design**
– Develop data infrastructure integrating protein sequence, structure, state, transition, and functional information.
– Construct benchmark tasks, datasets, splits, baselines, and evaluation metrics for multistate structure prediction and inverse folding.
– Develop state-aware annotation, representation-learning, and generative modelling methods for multistate protein design.
- **Selective Anticancer Peptide Design**
– Contribute to de novo peptide generation, selective anticancer screening, and cancer-type prediction.
– Design additional computational experiments and support evaluation of generated candidates.

Open-source Software and Data Infrastructure

- **KaleProtein (v0.1.0a1, PyPI)** – Founder and lead developer
 - Own the software architecture, implementation, testing, documentation, packaging, versioning, and release workflow.
 - Support protein sequence, structure, and small-molecule data through standardised model and dataset registration and loading interfaces.
 - Provide runnable DrugBAN and MapDiff workflows with the required datasets.
 - Maintain automated testing and continuous integration through GitHub Actions.
- **PyKale (480+ GitHub stars; 34K+ downloads)** – Core contributor and maintainer
 - Contribute to a multimodal machine-learning library included in the PyTorch Ecosystem.
 - Develop meta-learning modules, tests, documentation, and an end-to-end cancer multi-omics tutorial.
 - Support repository maintenance and a user-driven redesign informed by surveys and semi-structured interviews.
 - Fourth-largest contributor by commit count after its three co-founders.
- **MuSProt** – Lead developer
 - An open, million-scale multistate protein database.
 - Developed and deployed a Hugging Face-hosted multistate protein database and web platform.
 - Provide search, visualisation, and data download.
 - Released the reproducible dataset-construction workflow, including data collection and automated annotation methods.

Publications

Conference Papers

- [C1] **W. Fan**, L.M.R. Rizky, J. Zhang, C. Chen, H. Lu, K. Teh, D. Selvarajah, and S. Zhou, “Foundation-Model-boosted multimodal learning for fMRI-based neuropathic pain drug response prediction”, in *the 28th Int. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI, CORE A)*, pp. 238–248, Sep. 2025. DOI: 10.1007/978-3-032-05182-0_24.
- [C2] J. Zhang, X. Liu, W. Wu, S. Tabakhi, **W. Fan**, S. Zhou, K.L. Tee, T.S. Wong, and H. Lu, “Classifying the Stoichiometry of Virus-like Particles with Interpretable Machine Learning”, in *47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1–7, Jul. 2025. DOI: 10.1109/EMBC58623.2025.11254123.
- [C3] M.N.I. Suvon, P.C. Tripathi, **W. Fan**, S. Zhou, X. Liu, S. Alabed, V. Osmani, A. Swift, C. Chen, and H. Lu, “Multimodal Variational Autoencoder for Low-cost Cardiac Hemodynamics Instability Detection”, in *the 27th Int. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI, CORE A)*, pp. 296–306, Oct. 2024. DOI: 10.1007/978-3-031-72378-0_28.
- [C4] J. Wang, B. Chen, **W. Fan**, and Y. Liu, “Co-salient object detection via discriminative prototypes contrast”, in *the 2024 IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 7390–7394, Apr. 2024. DOI: 10.1109/ICASSP48485.2024.10446685.
- [C5] **W. Fan**, “Path planning for mobile robot in 3D environment based on ant colony algorithm”, in *the 2nd Int. Conf. on Artificial Intelligence and Information Systems (ICAIIIS)*, p. 012095, May. 2021. DOI: 10.1088/1742-6596/1982/1/012095.

Journal Papers

- [J1] **W. Fan**, L.M.R. Rizky, J. Zhang, C. Chen, H. Lu, K. Teh, D. Selvarajah, and S. Zhou, “Interpretable Foundation-model-boosted Multimodal Learning Facilitates Precision Medicine for Neuropathic Pain,” *npj Digital Medicine (IF: 15.1)*. Under review after major revision.
- [J2] P. Bai, X. Liu, **W. Fan**, T. Jiang, S. Alabed, W.K. Cheung, and H. Lu, “Geometry-aware Line Graph Transformer Pre-training for Molecular Property Prediction”, May. 2026, *IEEE Transactions on Neural Networks and Learning Systems (IF: 8.9)*.
- [J3] M. N. I. Suvon, S. Zhou, P. C. Tripathi, **W. Fan**, S. Alabed, B. Khanal, V. Osmani, A. J. Swift, C. (Cherise) Chen and H. Lu, “Multimodal Latent Fusion of ECG Leads for Early Assessment of Pulmonary Hypertension,” *IEEE Journal of Biomedical and Health Informatics (IF: 6.7)*. arXiv: 2503.13470, Under review after major revision.
- [J4] T. Payne, **W. Fan**, C. Chen, S. Zhou, V. Osmani, H. Lu, and O. Bandmann, “Risk of Parkinson’s disease following cholecystectomy: a prospective cohort study in the UK Biobank,” *npj Parkinson’s Disease (IF: 8.2)*. Under revision.

Preprints & Abstracts

- [P1] **W. Fan**, M. N. I. Suvon, S. Zhou, X. Liu, S. Alabed, V. Osmani, A. Swift, C. Chen, & H. Lu, “MeDSLIP: Medical Dual-Stream Language-Image Pre-training with Pathology-Anatomy Semantic Alignment”, *arXiv: 2403.10635*, 2024.

Leadership

- **Technical and research project leadership**
 - Translate biomedical questions into tractable machine-learning tasks and make key methodological decisions in the neuropathic pain and Parkinson's disease projects.
 - Coordinate technical delivery, collaborator discussions, action tracking, and follow-up.
 - Independently manage the end-to-end development and release of KaleProtein.
- **Bio-lab research assistant hackathon team – Team lead**
 - Built the main agentic system for laboratory management, experiment planning, parameter calculation, and real-time voice question answering.
 - Integrated supporting tools developed by other members of the team.
- **Self-driving protein and materials design lab hackathon team – Team lead**
 - Led a separate interdisciplinary team of biology, chemistry, mathematics, active-learning, and AI researchers.
 - Designed and implemented an agentic workflow for requirement-driven metal-binding protein design, evaluation, screening, and optimisation.
 - Led the integration, demonstration design, and technical presentation of the prototype.

Selected Grant Applications

- **UK Sovereign AI Strategic Assets proposal – Co-Investigator; approximately £1.27M**
 - Originated the core concept for an AI-ready benchmarking dataset spanning proteins and organic materials.
 - Led most of the expression-of-interest development and coordinated technical follow-up with academic and industry partners.
- **ARIA Activation Partner proposal – Co-applicant; £6.39M**
 - Co-developed the project concept, technical approach, and delivery plan.
- **UKRI Generative AI Hub proposals – Co-applicant; two proposals of £100K each**
 - Led the concept, technical planning, and application draft for one proposal.
 - Contributed to the conception of a second proposal.
- **Innovate UK Frontier AI Benchmarking Datasets proposals – Co-applicant; two proposals of £750K each**
 - Contributed technical concepts and benchmarking methodology.
- **ARIA x Encode AI Fellowship – Applicant; £115K**
 - Independently developed an application on AI-driven multistate protein design.

Dissemination, Training, and Professional Service

- **Selected Presentations**
 - Neuropathic pain drug-response prediction, MICCAI, Daejeon, September 2025.
 - Poster and one-minute pitch: “State-Aware Functional Annotations for Multi-State Design of Dynamic Proteins”, AI, Engineering Biology and Beyond, University of Bristol, January 2026.
 - Poster on multistate protein research, Prague Protein Spring, Prague, 2026.
 - “AI for Parkinson's Disease”, Sheffield AI Community Meeting, University of Sheffield, March 2024.
- **Training and Technical Support**
 - Produced the MOGONet cancer subtype prediction tutorial used in the PyKale tutorial at IEEE EMBC 2025.
 - Technical support, Fourth Workshop on Multimodal AI, London, June 2026.
 - Technical support, Mini-Hackathon on Multimodal AI at the Fourth Workshop on Multimodal AI, London, June 2026.
 - Technical support, Second Multimodal AI Community Forum, Sheffield, March 2026.
 - Technical support, Third Workshop on Multimodal AI, London, September 2025.
 - Technical support, Mini-Hackathon on Multimodal AI at the Third Workshop on Multimodal AI, London, September 2025.
- **Selected Journal Reviewer**
 - IEEE Transactions on Neural Networks and Learning Systems.
 - IEEE Journal of Biomedical and Health Informatics.
 - Computational and Structural Biotechnology Journal.
- **Selected Conference Reviewer**
 - International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2026.
 - MICCAI Workshop on Advancing Data Solutions in Medical Imaging AI (ADSMI), 2024.